PROJECT SYNOPSIS

PhishLens

Explainable Hybrid Phishing-URL Detection with Out-of-Template Evaluation

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Course	Design Project – II · Semester IV · 2024–25
Domain	Machine learning for security · phishing detection
Repository	github.com/harshacreate/PhishLens

Introduction

PhishLens is a local, explainable phishing-URL analyser with an honest evaluation of how well it generalises beyond its training data.

Problem Statement

URL classifiers report high accuracy on in-distribution test sets but rarely explain decisions or measure performance on unseen attack styles.

Objectives

- 16 interpretable features and rule signals
- Explainable logistic regression with per-feature contributions
- Four-model comparison with cross-validation
- Pre-registered out-of-template probe evaluation
- Typosquat/combosquat signals and hybrid scoring
- Authorization-gated, SSRF-safe passive enrichment

Methodology

Synthetic seeded dataset; shared feature extractor for training and serving; standardised logistic regression; comparison via scikit-learn; hand-written development and hold-out probe sets; hybrid floor for high-severity signals; Flask API, Docker and pytest.

Tools

Python, Flask, scikit-learn, numpy, dnspython, whois, OpenSSL, Docker, pytest, GitHub Actions.

Expected Outcomes

- Working analyser with single, e-mail and batch modes
- Quantified in-distribution and out-of-template performance
- Security review of network features
- Report, presentation and demonstration

References

- Ma et al., KDD 2009; Sahingoz et al., ESWA 2019
- Kintis et al., CCS 2017 (combosquatting)
- Geirhos et al., Nature Machine Intelligence 2020 (shortcut learning)
- OWASP SSRF Prevention Cheat Sheet